Exercise Sheet

Lecture 15

Perturbation Theory for Linear Operators

Recommended optional route: 7 recommended + 3 extensions

This sheet accompanies the optional, unassessed capstone. Work Exercises 15.1–15.7 for the recommended optional route (typically 60–75 minutes). Exercises 15.8–15.10 are extensions; allow roughly 100–125 minutes for all ten. Throughout, $H(\varepsilon) = H_0 + \varepsilon V$ with H_0 and V self-adjoint on a finite-dimensional complex inner-product space and $\varepsilon \in \mathbb{R}$, and denominators are written with the level of interest first.
In the infinite-dimensional extensions, $\mathcal{H}$ is complex, $\mathcal{B}(\mathcal{H})$ denotes bounded operators, and every unbounded operator is understood together with its stated domain.

Exercise 15.1 (computational) Let $H_0 = \operatorname{diag}(1, 4)$ and $V = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Find the first- and second-order shifts of both levels, and compare the second-order prediction with the exact eigenvalues at $\varepsilon = 0.1$.

Answer. $E^{(1)} = 0$ for both; $E_1^{(2)} = -\frac{1}{3}$, $E_2^{(2)} = +\frac{1}{3}$. At $\varepsilon = 0.1$: predicted 0.996667 and 4.003333, exact 0.996670 and 4.003330.

Exercise 15.2 (computational) For $H_0 = \operatorname{diag}(0, 2, 5)$ and

$$V = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

compute all three first- and second-order shifts, and verify the two checks $\sum_n E_n^{(1)} = \operatorname{tr} V$ and $\sum_n E_n^{(2)} = 0$.

Answer. $E^{(1)} = (1, -1, 0)$, summing to $0 = \operatorname{tr} V$; $E^{(2)} = \left(-\frac{1}{2}, -\frac{5}{6}, \frac{4}{3}\right)$, summing to 0.

Exercise 15.3 (computational) For the three-level system of the lecture, $H_0 = \operatorname{diag}(0, 1, 3)$ with V having rows $(2, 1, 1)$, $(1, 1, 1)$, $(1, 1, -3)$, compute the first-order correction to the *top* eigenvector and evaluate it at $\varepsilon = 0.1$.

Answer. $|3^{(1)}\rangle = \frac{1}{3}|1\rangle + \frac{1}{2}|2\rangle$, giving $(0.0333, 0.05, 1)$ at $\varepsilon = 0.1$; in the same normalization the exact vector is approximately $(0.0424, 0.0647, 1)$.

Exercise 15.4 (computational) Let $H_0 = \operatorname{diag}(0, 1, 3)$ and $V = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. Compute $\|V\|$, and find the range of $|\varepsilon|$ for which the three *closed* disks of the spectral stability theorem are pairwise disjoint. State the supremal threshold.

Answer. $\|V\| = 1$; the closed disks are pairwise disjoint for $|\varepsilon| < \frac{1}{2}$, whose supremal threshold is $\frac{1}{2}$ (not an admissible largest value).

Exercise 15.5 (proof) *(Hellmann–Feynman.)* Let $|n(\varepsilon)\rangle$ be a normalized eigenvector of $H(\varepsilon)$ depending differentiably on ε, with eigenvalue $E_n(\varepsilon)$. Prove that

$$\frac{dE_n}{d\varepsilon} = \langle n(\varepsilon)| V |n(\varepsilon)\rangle,$$

and deduce the first-order formula $E_n^{(1)} = \langle n| V |n\rangle$.

Hint. Write $E_n = \langle n(\varepsilon)| H(\varepsilon) |n(\varepsilon)\rangle$ and differentiate, using $\langle n(\varepsilon) \mid n(\varepsilon)\rangle = 1$.

Exercise 15.6 (proof) Assume H_0 has simple eigenvalues. Prove that $\sum_n E_n^{(2)} = 0$, and that the second-order shift of the *lowest* level is always ≤ 0 while that of the highest is always ≥ 0.

Exercise 15.7 (computational) Let $H_0 = \operatorname{diag}(2, 2, 6)$ and

$$V = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Explain why the non-degenerate formulas cannot be used, then find the first-order branches of the degenerate level and their zeroth-order states. Finally explain why PVP does not guarantee splitting or a unique basis in general, using $H_0 = \operatorname{diag}(1, 1, 4)$ and $V = I$.

Answer. $\langle 1| V |2\rangle = 3 \ne 0$ with $E_1 - E_2 = 0$. Compressing, $PVP = \left(\begin{smallmatrix} 1 & 3 \\ 3 & 1 \end{smallmatrix}\right)$ has distinct eigenvalues $4, -2$, so the branches are $2+4\varepsilon$ and $2-2\varepsilon$, on $\frac{1}{\sqrt{2}}(|1\rangle \pm |2\rangle)$. For the counterexample $PVP = P$, both slopes are 1: no first-order splitting and no preferred basis.

Optional extensions/outlook: Exercises 15.8–15.10

Optional extension: complex branch points

Exercise 15.8 ($\star$) For $\Delta > 0$ consider the two-level family

$$H(\varepsilon) = \begin{pmatrix} \Delta/2 & \varepsilon \\ \varepsilon & -\Delta/2 \end{pmatrix}.$$

(a) Find the exact eigenvalues. (b) Expand the upper one to order ε^2 and check it against second-order perturbation theory. (c) Regarding ε as a complex variable, find the radius of convergence of that expansion, and say what obstructs it.

Hint. For (c), ask where the square root in (a) has a branch point.

Answer. (a) $\pm\sqrt{\Delta^2/4 + \varepsilon^2}$. (b) $\Delta/2 + \varepsilon^2/\Delta$, matching $E_+^{(2)} = 1/\Delta$. (c) Radius $\Delta/2$, obstructed by the branch points at $\varepsilon = \pm i\Delta/2$, where the two levels would collide.

Optional extension/outlook: infinite dimensions

Exercise 15.9 (computational) Let D be the diagonal operator $De_n = \frac{1}{n} e_n$ on ℓ^2 and let $z = -1$. Compute $\|(D - z\,\mathrm{id})^{-1}\|$, and determine the range of $|\varepsilon|$ for which the resolvent expansion is *guaranteed uniformly using only the stated norm data*, together with its supremal radius, $(D + \varepsilon V - z\,\mathrm{id})^{-1} = \sum_{k\ge 0} (-\varepsilon)^k R_0(z) \big(V R_0(z)\big)^k$ for a bounded self-adjoint V with $\|V\| = 2$. Show that the uniform half-radius is sharp for $V = 2I$, whereas $V = 2P_1$ permits the larger range $|\varepsilon| < 1$, where P_1 projects onto $\operatorname{span}(e_1)$.

Answer. $\operatorname{spec}(D) = \{1/n\} \cup \{0\}$, so $\operatorname{dist}(-1, \operatorname{spec} D) = 1$ and $\|R_0(-1)\| = 1$; $|\varepsilon| < \frac{1}{2}$ is the uniform sufficient range, with supremal radius $\frac{1}{2}$. For $V = 2I$, $\|V R_0\| = 2$ and the operator-norm series fails at the boundary $|\varepsilon| = \frac{1}{2}$; for $V = 2P_1$, $\|V R_0\| = 1$, so it converges only for the strict range $|\varepsilon| < 1$, of supremal radius 1.

Exercise 15.10 ($\star$) The Bender–Wu analysis of the quartic anharmonic oscillator $H(\varepsilon) = \frac{1}{2}(\hat{p}^2 + \hat{x}^2) + \varepsilon \hat{x}^4$ finds divergent large-order behaviour and complex branch points accumulating at $\varepsilon = 0$. Explain why this gives zero convergence radius yet permits asymptotically useful low-order accuracy. Why is negative coupling only physical motivation, not a spectral proof without a chosen self-adjoint realization? Contrast the finite-dimensional simple-level case.

Hint. Use the cited branch points for convergence, then identify the domain and boundary data missing from the formal differential expression.